

Unmasking Political Question Evasions using Transformer-Based Models

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I. INTRODUCTION

The objective of this project is to determine whether a politician answer a question directly or evades it. Political interviews are often filled with indirect, unrelated or partial answers. Therefore in order to provide transparency and accountability in public communications, we developed a model for detecting these elusive answers. We are treating this task as a question/answer classification task, following the structure of the SemEval CLARITY dataset which categorizes responses as direct, partial or evasive. Our current pipeline starts by performing EDA operations on our data before sending it to be preprocessed. Following preprocessing we move on to feature extraction where we extracted several features from the dataset in order to feed them to our models. Afterwards we fed the data as well as the features to multiple ML models from simple models such as SVM and Random Forest to transformer models such as RoBERTa. Finally we evaluated our results by calculating their macro f1, macro precision and macro recall as well as their precision recall curves. By utilizing this pipeline we were able to achieve satisfactory results on par with the actual models used "I Never Said That": A dataset, taxonomy and baselines on response clarity classification.[1]

II. RELATED WORK

A. Political Question Evasion Detection

Political question evasion has been extensively studied in political science and discourse analysis, particularly in the context of political interviews where public figures may avoid directly answering challenging questions. Evasive answering behavior has been characterized through strategies such as ambiguity, topic shifting, partial answers, and indirect responses. A major step toward formalizing this phenomenon in computational settings is the CLARITY dataset introduced by [1], which provides a large-scale, manually annotated corpus of political question-answer pairs and categorizes responses into *Clear Reply*, *Ambivalent*, and *Clear Non-Reply*. This work establishes political question evasion detection as a supervised classification problem and offers standardized evaluation benchmarks.

Earlier studies on evasive answering primarily relied on qualitative analysis and manual annotation. Research in related domains, such as financial question-answer sessions, proposed discourse-driven taxonomies of evasive behavior and identified linguistic characteristics commonly associated with evasion, including increased hedging, indirect phrasing, tense shifts, and reduced semantic alignment between questions and answers [2]. While these studies provided important theoretical foundations and insights into evasive strategies, their reliance on manual analysis limited their scalability and applicability to large datasets.

From a computational perspective, prior approaches have modeled evasion detection using classical supervised learning methods. These methods typically rely on surface-level linguistic cues, lexical overlap, answer length, and similarity measures between question and answer texts. The dataset[1] demonstrates that such features are effective at capturing certain forms of evasive behavior, particularly explicit non-answers, but often struggle with subtle semantic distinctions and cases where responses are partially relevant. Moreover, class imbalance—where ambiguous answers dominate the dataset—poses additional challenges for traditional classifiers, motivating the use of evaluation metrics such as macro-F1.

B. Transformer-based Text Classification

Recent advances in natural language processing have led to the widespread adoption of transformer-based models for text classification. By capturing long-range dependencies and contextual semantics, transformers are particularly well-suited for political question-answer settings, where evasive responses often rely on subtle semantic shifts rather than explicit lexical cues [1][2].

Following the formulation of [1], transformer-based methods model political question evasion as a sequence-pair classification task by jointly encoding questions and answers. This approach enables learning semantic alignment and topic drift directly from context and has been shown to outperform classical baselines, especially on ambiguous cases [1].

Prior work further highlights the importance of discourse structure and external context. Majumder et al. [3] demonstrate that incorporating discourse patterns and knowledge grounding improves dialog understanding. As a result, hybrid approaches that combine transformer representations with interpretable

linguistic or grounding-based features have been proposed to improve robustness and interpretability [1][4].

C. Baseline and State-of-the-Art Approaches

Baseline methods for political question evasion detection often employ classical machine learning models trained on handcrafted features and TF-IDF representations, as demonstrated in prior computational studies on evasive answering behavior [2]. Common classifiers include Naive Bayes, Logistic Regression, Support Vector Machines, and tree-based models such as Random Forests. These approaches provide interpretable and computationally efficient baselines and remain competitive in many settings.

a) *Counterfactuals of Counterfactuals*: We also consider model robustness through counterfactual explanation methods. Classical counterfactuals search for the smallest perturbation that flips the prediction, offering insight into black-box decision boundaries. The referenced work proposes evaluating counterfactual *quality* by generating a second counterfactual from the first [5].

III. METHODOLOGY

A. Dataset Description

1) *Dataset Overview*: The dataset[1] is obtained from the QEvasion benchmark, which contains manually annotated political interview transcripts. The dataset consists of political interview question-answer pairs annotated for the task of question evasion detection. Each instance includes an interview question, the corresponding answer given by a public figure, and a categorical label indicating the level of evasiveness. The classification task is formulated as a three-class problem with the following labels: *Clear Reply*, *Ambivalent*, and *Clear Non-Reply*.

The dataset is divided into training and test splits. The training set is used for model training and hyperparameter optimization, while the held-out test set is reserved exclusively for final evaluation.

2) *Label Distribution*: An exploratory analysis of the label distribution reveals a class imbalance across the three categories, with *Clear Reply* being the most frequent class. To address potential bias introduced by this imbalance, we employ macro-averaged evaluation metrics throughout all experiments, ensuring that each class contributes equally to the reported performance scores.

B. Problem Formulation

Given an interview question q and its corresponding answer a , the goal is to predict an evasiveness label $y \in \{0, 1, 2\}$, representing *Clear Reply*, *Ambivalent*, and *Clear Non-Reply*, respectively. This is modeled as a multi-class text classification problem. The input consists of paired textual sequences, and the output is a single class prediction.

C. Text Preprocessing

1) *Cleaning and Normalization*: We apply a structured preprocessing pipeline to support both classical and hybrid modeling approaches. The text is lowercased and cleaned of punctuation, while stopword removal is applied selectively by preserving negation cues and personal pronouns, which are known to be informative for evasiveness detection. Samples with missing or undefined labels are removed from the dataset.

In addition, lemmatization and stemming are applied for feature-based models, while the raw text is preserved for transformer-based models to avoid loss of semantic information.

2) *Tokenization Strategy*: For transformer-based models, we employ the `RobertaTokenizerFast` tokenizer associated with the *roberta-base* model. Each input is constructed as a paired sequence consisting of the interview question and its corresponding answer. Tokenization is performed with truncation and padding to a maximum sequence length of 256 tokens to ensure uniform input size. This allows the model to jointly reason over both the question and the answer.

D. Feature Engineering

1) *Linguistic Features*: In addition to raw text inputs, auxiliary linguistic features are extracted in earlier stages of the pipeline, including syntactic and lexical statistics. These features are explored in classical and hybrid baselines; however, the transformer-based model relies primarily on contextual embeddings learned directly from the text.[2]

2) *Statistical Features*: We extract a diverse set of statistical and structural features designed to capture surface-level and discourse-level characteristics of evasive answers. These include length-based features such as question and answer word counts, sentence counts, and answer-to-question length ratios.

To capture semantic alignment, we compute lexical overlap metrics including Jaccard similarity, content-word overlap, and named-entity overlap between questions and answers. In addition, contextual semantic similarity is explored using SBERT-based cosine similarity scores.

Discourse-level indicators such as tense distributions, negation ratios, and hedge marker frequencies are extracted to model uncertainty, deflection, and temporal avoidance strategies commonly observed in evasive responses. These features are primarily used in classical and hybrid baselines and are not directly injected into the transformer model.

E. Baseline Models

To establish reference performance levels, we implement a diverse set of baseline models using handcrafted linguistic and statistical features. These baselines aim to quantify the predictive power of explicit surface-level, syntactic, and discourse-driven cues, in contrast to contextual representations learned by transformer-based models.

Classical baselines operate on fixed feature representations extracted from question-answer pairs and are trained using standard supervised learning algorithms:

- Logistic Regression

- Linear Support Vector Machines
- Naive Bayes (Gaussian)
- Decision Trees

These models rely on compact feature sets including lexical overlap (e.g., Jaccard similarity), structural indicators (e.g., multiple-question flags), and basic length-based statistics.

Advanced and hybrid baselines extend classical approaches by incorporating richer linguistic and semantic features, while remaining outside end-to-end transformer fine-tuning:

- Random Forest classifiers trained on aggregated linguistic, syntactic, and semantic features
- Gradient-boosted decision trees (XGBoost) leveraging discourse-level indicators such as hedge markers, negation cues, tense distributions, and contextual similarity scores
- Feature-augmented classifiers incorporating SBERT-based semantic similarity measures

All baseline models are evaluated using macro-averaged metrics to ensure robustness against class imbalance. These baselines serve as strong reference points for assessing the gains provided by transformer-based architectures.

F. Transformer-Based Model

1) *Model Architecture*: Our primary model is based on *RoBERTa-base*, a pretrained transformer encoder with 12 layers and a hidden dimension of 768. A task-specific classification head is added on top of the contextual representation of the input sequence, derived from the special classification token. The model is fine-tuned end-to-end for the three-class classification task.

2) *Input Representation*: The model input is formed by concatenating the interview question and the interview answer as a single sequence pair. This design allows the transformer to capture interactions between the question and the answer, which is critical for detecting evasive responses that depend on contextual alignment.

3) *Training Procedure*: The model is trained using the Hugging Face *Trainer* API. Training is performed with mixed-precision (FP16) to improve computational efficiency. The dataset is split into training and validation subsets, and evaluation is conducted at the end of each epoch. The best model checkpoint is selected based on the validation macro F1-score.

G. Hyperparameter Optimization

1) *Search Space*: Hyperparameter optimization is conducted using Optuna. The search space includes the learning rate, batch size, number of training epochs, and weight decay. Learning rates are sampled on a logarithmic scale, while batch sizes and epoch counts are chosen from discrete ranges.

2) *Optimization Strategy*: The optimization objective is to maximize the macro F1-score on the validation set. Multiple trials are executed, and poorly performing configurations are pruned early. The best-performing hyperparameter configuration is then used to retrain the model on the combined training and validation data before final evaluation on the test set.

IV. RESULTS

A. Baseline Performance

This section presents the results obtained from the classical baseline models trained on handcrafted feature sets. We analyze both text-based TF-IDF representations and numeric handcrafted features, highlighting strengths and limitations observed during experimentation.

1) *TF-IDF-Based Text Models*: Among the TF-IDF-based models, Naive Bayes and Linear SVM produced stable and competitive results. Naive Bayes achieved reasonable macro-F1 scores, particularly performing well on the majority class, but showed limited recall for minority classes. This behavior is expected due to Naive Bayes' conditional independence assumption, which assumes that individual TF-IDF features are independent given the class label. In high-dimensional and sparse text representations, this assumption is often violated, as word co-occurrence patterns and contextual dependencies play an important role in distinguishing subtle semantic differences. As a result, Naive Bayes tends to overemphasize frequent lexical cues from the majority class while failing to capture discriminative feature interactions necessary for correctly identifying minority class instances, leading to reduced recall.

Linear SVM consistently outperformed Naive Bayes across most evaluation metrics. Its ability to handle high-dimensional sparse TF-IDF features resulted in improved class separation and higher macro-F1 scores, especially for minority classes. Overall, Linear SVM emerged as the strongest classical text-based baseline in our experiments.

Logistic Regression was also evaluated on TF-IDF features. However, due to the high dimensionality of the feature space, training was significantly slower compared to the other models. After simplifying the TF-IDF configuration and adjusting the solver settings, Logistic Regression achieved performance levels comparable to those of Linear SVM. The simplification process involved reducing the feature dimensionality by limiting the vocabulary size and tuning regularization parameters, which significantly decreased training time and improved numerical stability.

Despite these improvements, Logistic Regression still exhibited slightly inferior macro-F1 performance compared to Linear SVM. This difference can be attributed to Linear SVM's margin-based optimization objective, which is particularly effective in high-dimensional and sparse feature spaces such as TF-IDF representations. Linear SVM is able to construct more robust decision boundaries under these conditions, whereas Logistic Regression relies on probabilistic modeling that may be more sensitive to feature sparsity and class imbalance.

Given its higher computational cost during training and inference, combined with performance that does not surpass that of Linear SVM, Logistic Regression was not selected as the primary linear baseline model. Instead, Linear SVM was preferred due to its superior efficiency-to-performance trade-off and more consistent behavior across all classes.

2) *Random Forest on Handcrafted Numeric Features*: The Random Forest model was trained using only handcrafted

numeric features, including length-based statistics, similarity scores, and rule-based indicators. While its overall performance was lower than the TF-IDF-based text models, Random Forest demonstrated the ability to capture non-linear relationships among numeric features.

The results indicate that numeric handcrafted features alone provide useful but limited information compared to text-based representations. Feature importance analysis showed that similarity-related features and length ratios contributed most to the model’s decisions, whereas simpler rule-based features had lower impact.

3) *Overall Observations:* Across all classical baselines, text-based TF-IDF representations clearly outperformed numeric-only handcrafted features. Among the evaluated classifiers, Linear SVM achieved the best balance between performance and computational efficiency. These findings suggest that lexical and contextual information extracted via TF-IDF plays a crucial role in the clarity classification task, while numeric handcrafted features serve as a complementary but weaker signal.

Despite their limitations, classical models provide strong and interpretable baselines, establishing a solid reference point for comparison with more advanced neural approaches in subsequent experiments.

B. Advanced Machine Learning Algorithms

1) *Random Forest Results:* The Random Forest model was trained using handcrafted numeric features such as length statistics, similarity scores, and rule-based indicators. On the validation set, the untuned model achieved a macro-F1 score of 0.518, while hyperparameter tuning improved the performance slightly to 0.530. The model performed best on the majority *Ambivalent* class, whereas performance on minority classes remained limited. These results indicate that numeric handcrafted features provide useful but incomplete information for the task.

2) *XGBoost Results:* XGBoost was evaluated on the same handcrafted numeric feature set. The model achieved a macro-F1 score of 0.508 on the validation set, demonstrating strong performance on the majority class but substantially weaker results for minority classes.

Despite its higher modeling capacity and ability to capture complex non-linear relationships, XGBoost did not significantly outperform Random Forest. This suggests that the primary performance bottleneck lies in the feature representation rather than the expressive power of the classifier itself. In other words, the handcrafted numeric features appear to provide limited discriminative information for separating subtle evasiveness patterns, particularly for minority classes.

Simply increasing model complexity does not yield meaningful gains when the input features fail to adequately encode semantic alignment, discourse structure, and pragmatic cues present in question–answer pairs. Therefore, further performance improvements are more likely to come from enhancing feature representations rather than employing more complex tree-based models.

Potential directions include incorporating richer semantic features (e.g., contextual sentence embeddings), discourse-level representations, or hybrid approaches that combine numeric features with learned text representations. This observation motivates the transition toward transformer-based models, which learn task-specific feature representations directly from raw text and are better suited to capturing nuanced forms of political question evasion.

3) *Neural Network:* The Neural Network model was evaluated by using handcrafted numerical features consisting of 28 features including length-based statistics, lexical overlap measures, punctuation indicators, and discourse-level signals. The model achieved a macro-F1 score of 0.564 which was higher than both previous advanced models. To be more specific, our model performed best on the Ambivalent class (F1 = 0.64), while showed a better recall for the Clear Non-Reply class (recall = 0.69). Model’s performance on the Clear Reply was highly lower than others (F1 = 0.47). All in all, overall class balance had a noticeable advancement compared to tree-based models. Hyperparameter tuning such as network depth, hidden layer size, dropout rate, and learning rate, improved performance. Nevertheless, the results demonstrated us even though the neural networks are able to capture non-linear patterns better, overall performance is limited by the handcrafted feature set.

TABLE I
CLASSIFICATION REPORT FOR THE NEURAL NETWORK (PYTORCH MLP)
MODEL ON THE DEVELOPMENT SET

Class	Precision	Recall	F1-Score	Support
Clear Reply	0.42	0.53	0.47	105
Ambivalent	0.73	0.57	0.64	204
Clear Non-Reply	0.50	0.69	0.58	36
Accuracy			0.57	345
Macro Avg	0.55	0.60	0.56	345
Weighted Avg	0.61	0.57	0.58	345

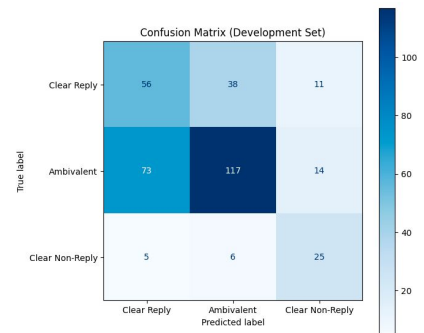


Fig. 1. Confusion Matrix for Neural Network

C. Encoder Transformer Model Performance

In this experiment, we evaluate a transformer-based encoder model without incorporating any external grounding information. The goal is to assess how well a pretrained contextual language model can capture evasiveness patterns solely from the interview question–answer pairs.

We fine-tune *RoBERTa-base*, a pretrained encoder-only transformer, on the three-class evasiveness classification task. Each input consists of a paired sequence of the interview question and the corresponding answer, tokenized using the RoBERTa tokenizer with a maximum sequence length of 256 tokens. No handcrafted features or external grounding signals are provided to the model.

Hyperparameter optimization is performed using Optuna, with the objective of maximizing the macro F1-score on the validation set. The search space includes the learning rate, batch size, number of training epochs, and weight decay. After selecting the best hyperparameter configuration, the model is retrained on the combined training and validation data and evaluated on the held-out test set.

On the test set, the without grounding RoBERTa model achieves an accuracy of 0.67 and a macro F1-score of 0.57. The macro precision and macro recall are 0.56 and 0.60, respectively, indicating a relatively balanced performance across classes despite the underlying class imbalance.

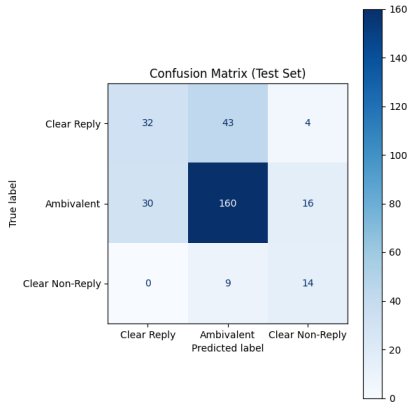


Fig. 2. Confusion Matrix of the Encoder Model Without Grounding

Figure 2 presents the confusion matrix for the without grounding encoder model. The model performs strongest on the majority *Ambivalent* class, while confusion is most frequently observed between *Clear Reply* and *Ambivalent*. Performance on the *Clear Non-Reply* class remains more limited, which is consistent with its lower representation in the dataset.

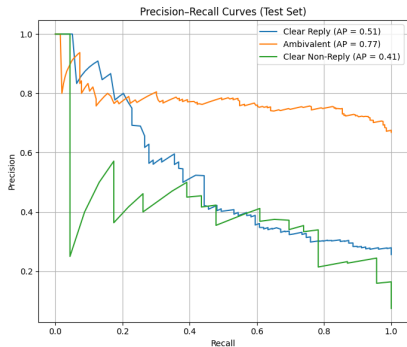


Fig. 3. Confusion Matrix of the Encoder Model Without Grounding

Figure 3 illustrates the precision–recall curves for each class. The *Ambivalent* class exhibits the highest average precision, while *Clear Non-Reply* remains the most challenging class to detect reliably. These observations highlight the difficulty of identifying subtle forms of non-response without external contextual grounding.

Overall, the results demonstrate that a pretrained encoder-only transformer can capture meaningful evasiveness patterns directly from text. However, the performance gap between majority and minority classes suggests that additional contextual signals or grounding information may further improve robustness, motivating the grounded encoder experiments presented in the next section.

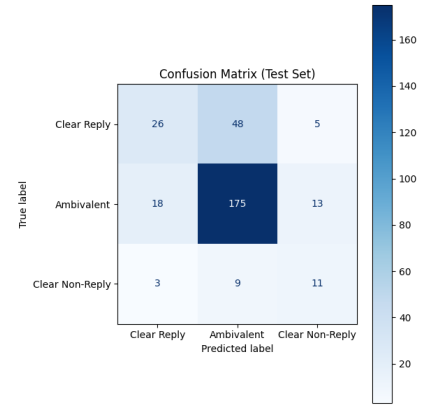


Fig. 4. Confusion Matrix of the Encoder Model With Grounding

The numeric-grounding-enhanced RoBERTa model achieves an overall accuracy of 0.688 with a macro F1 score of 0.545 on the test set. As shown in Figure 4, the model demonstrates strong performance on the *Ambivalent* class, correctly identifying the majority of ambivalent instances, while still exhibiting confusion between *Clear Reply* and *Ambivalent*. This behavior is expected given the semantic proximity of these classes and reflects relatively conservative decision boundaries.

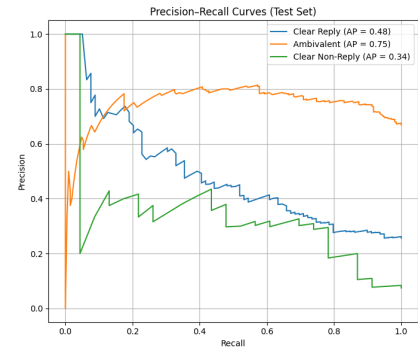


Fig. 5. Confusion Matrix of the Encoder Model With Grounding

The precision–recall curves in Figure 5 further support this observation. The *Ambivalent* class achieves a comparatively high average precision (AP = 0.75), indicating that numeric

grounding signals effectively capture ambiguity-related patterns. In contrast, *Clear Reply* (AP = 0.48) and *Clear Non-Reply* (AP = 0.34) remain more challenging to distinguish, suggesting that while grounding alignment improves ambiguity detection, fine-grained separation between clear response types remains difficult. Overall, the macro average precision of 0.524 confirms that numeric grounding contributes meaningful but class-dependent improvements.

Compared to the direct clarity RoBERTa-base baseline reported in Table[1], which achieves a macro F1 score of approximately 0.53, the proposed numeric-grounding-enhanced model demonstrates a modest yet consistent improvement in macro-level performance. The gain is primarily driven by improved recall-precision balance on the *Ambivalent* class, aligning with the intended role of grounding features such as question-grounding and answer-grounding cosine similarities.

Although the overall improvement remains incremental, this comparison indicates that explicit grounding alignment signals complement RoBERTa’s linguistic representations, particularly for ambiguity-sensitive classification. These findings suggest that grounding is most beneficial for tasks requiring reasoning over uncertainty and partial alignment rather than relying solely on surface-level textual cues.

D. Decoder Transformer Model Performance

In decoder models, we have selected two main models which are "Llama-3.1-8B-Instruct" and "Llama-3.3-70B-Instruct". We finetuned these models using by LoRA from tinkr platform and we only used default question, answer and label features for finetuning. Before training, we splitted training set into new training set and validation set with percentages of 80% and 20% respectively. We used validation test to compare different hypertuned versions among themselves and we used test set for final evalaution.

1) *Llama-3.1-8B-Instruct Performance*: On the test set best Llama 8B model’s evaluation scores were the following. Macro F1 score: 0.6132, Macro Precision: 0.72, Macro Recall: 0.57. You can see more detailed analyses at the table and the graph bellow. As you can see from the table, Macro F1 score is lower than Weighted F1 score. This was expected due to the class imbalance across the labels. We have relatively high precision and low recall which indicates that model’s predictions are usually correct but it might not be able to catch all the instances of a class. For instance, if model predicts "Clear reply" it is usually correct but model isn’t able to detect majority of "Clear reply" true values. The state of art models Llama 3.1 8B that fine tuned on this dataset had 0.504 Weighted F1 score on the test set [1] while our model had 0.71 weighted f1 score. This shows that our finetuned Llama behaves significantly better than the state of art. The main reason can be caused by our model type selection because we fine tuned a "Instruct" Llama 8B which is better for few shot classification task.

TABLE II
CLASSIFICATION REPORT FOR THE BEST PERFORMING LLAMA 8B MODEL

Class	Precision	Recall	F1-Score	Support
Ambivalent	0.76	0.86	0.80	206
Clear Non-Reply	0.82	0.39	0.53	23
Clear Reply	0.57	0.46	0.51	79
Accuracy			0.72	308
Macro Avg	0.72	0.57	0.61	308
Weighted Avg	0.71	0.72	0.71	308

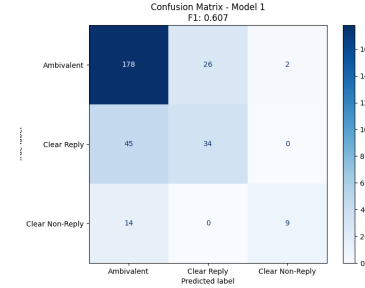


Fig. 6. Confusion Matrix for Llama 8B Instruct

2) *Llama-3.1-70B-Instruct Performance*: On the test set best Llama 70B model’s evaluation scores were the following. Macro F1 score: 0.48, Macro Precision: 0.81, Macro Recall: 0.45. You can see more detailed analyses at the table and the graph bellow. As you can see from the table, the same macro precision and macro recall gap that Llama 8B suffers can be observed here as well in a bigger magnitude. This indicates that while finetuning Llama 70B learns the majority label and ignore others. You can see this from its poor recall score at minority labels of Clear Non-Reply and Clear Reply which have recall score of 0.21 and 0.16 respectively. This output had contradicted our assumption that bigger models, perform better by getting a worse result macro f1 score than Llama 8B model. On the other hand, the state of art Llama 70b that fine tuned by the creators of this dataset which got an weighted f1 score of 0.75 [1] while our model got a weighted f1 score 0.71 which seems close.

TABLE III
CLASSIFICATION REPORT FOR THE BEST PERFORMING LLAMA 70B MODEL

Class	Precision	Recall	F1-Score	Support
Ambivalent	0.71	0.98	0.82	206
Clear Non-Reply	1.00	0.22	0.36	23
Clear Reply	0.72	0.16	0.27	79
Accuracy			0.71	308
Macro Avg	0.81	0.45	0.48	308
Weighted Avg	0.73	0.71	0.64	308

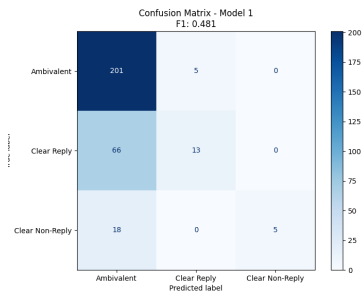


Fig. 7. Confusion Matrix for Llama 70B Instruct

3) *Details of Hyperparameter tuning and Experiments:* In the hyperparameter tuning, we used two main hyperparameters such as number of epochs and batch size. By decreasing and increasing epoch size we prevent overfitting and underfitting respectively. We use batch size for controlling the training speed with a trade off from the model performance. On the other hand, we didn't try any different values for learning rate and LoRA rank. Instead we used tinkers's default value for LoRA which is 32 and Tinker's recommended learning rate which changes across different models. We done several experiments that we didn't use for decoder fine tuning. Firstly, we have tried oversampling in order to decrease label imbalance at the dataset but it back fired because we feed duplicates of the same minority class instances to the model which resulted in poor accuracy. Then, we tried using a more sophisticated prompt instead of just giving the answer and question but it didn't improve performance.

V. DISCUSSION

A. Impact of Dataset Characteristics

The characteristics of the CLARITY dataset play a central role in shaping model performance across all experiments. Most notably, the dataset exhibits a pronounced class imbalance, with the Ambivalent class dominating the distribution, while Clear Non-Reply remains severely underrepresented. This imbalance consistently influenced model behavior, particularly in terms of recall for minority classes, and motivated our use of macro averaged evaluation metrics throughout the study. Aside from this the annotators who assigned the labels agreed only 60 percent of the time, meaning even the ground truth given does not represent the whole truth. Meaning rather than having golden labels we have a silver ground truth which also negatively affected the results.

Models trained on surface level features or purely numeric representations struggled to generalize beyond majority class patterns. In contrast, text-based models, especially transformer-based architectures, were better able to mitigate some effects of imbalance by leveraging contextual semantic information. However, even the strongest models showed persistent confusion between Clear Reply and Ambivalent, reflecting the inherent semantic proximity of these categories in real political discourse. These observations suggest that a significant portion of the difficulty in political question evasion

detection stems not only from model limitations, but also from the intrinsic ambiguity present in the dataset annotations.

B. Model Strengths and Weaknesses

The experimental results highlight clear trade-offs between classical, encoder-based, and decoder-based approaches. Classical models such as Linear SVM demonstrated strong performance relative to their simplicity, particularly when using TF-IDF representations. Their strengths lie in computational efficiency and interpretability; however, they lack the capacity to model deeper semantic relationships between questions and answers.

Advanced machine learning models operating on handcrafted numeric features, including Random Forest, XGBoost, and neural networks, captured non-linear interactions more effectively than classical baselines. Among these, the neural network achieved the best macro-F1 score, indicating improved class balance. Nevertheless, overall performance remained constrained by the expressiveness of the feature set, suggesting that handcrafted features alone are insufficient for capturing subtle evasive strategies.

Encoder-based transformer models (RoBERTa) exhibited improved robustness and class balance by jointly encoding question-answer pairs. The grounding-enhanced variant demonstrated a slightly worse performance across the board. Establishing a deeper relationship between words seems to have caused the models to be slightly more biased towards ambivalent and clear non-reply. However, performance on Clear Non-Reply still remained limited, highlighting the difficulty of detecting indirect or topic-shifting responses without broader discourse context.

Decoder-based models, particularly the fine-tuned Llama-3.1-8B-Instruct model, achieved the strongest overall performance. Its high macro precision indicates reliable predictions when a class is identified, though recall remained uneven across classes. The larger Llama-3.1-70B-Instruct model did not yield proportional gains and instead exhibited overconfidence in the majority class, underscoring that increased model size alone does not guarantee better macro-level performance under class imbalance.

C. Comparison with Related Work

We compared our work with the work of writers of "I never said that" [1]. Unlike us they didn't use any baseline or advanced machine learning models so we couldn't compare our machine learning models with them. Instead, they used encoder and decoder models and we have achieved similar results with the for both of them as we have stated in the result section. Both our work and their work had limited f1 scores due to the class imbalance which was expected.

D. Limitations

Despite encouraging results, the proposed system has several limitations. First, class imbalance remains a fundamental challenge, affecting recall for minority classes across nearly all models. Second, grounding signals were limited to numeric similarity measures and did not incorporate external

knowledge sources or broader discourse context, which may be essential for detecting sophisticated evasive strategies.

Additionally, decoder-based models were fine-tuned using default LoRA configurations and limited hyperparameter exploration. More extensive tuning or alternative adaptation strategies could further improve performance. Finally, the evaluation relies on a single dataset, and generalization to other political domains or languages remains untested.

E. Potential Improvements

Future work could explore several direction to address the problems and limitations we faced. Richer grounding sources with more detailed information about politics and presidents can improve model accuracy. Using a more balanced dataset or sampling a more balance subset from this dataset could be more useful for unbiased training.

VI. CONCLUSION

In this problem we classified question answer pairs into three labels. We used preprocessing and feature extraction in order to increase model accuracy especially for classical and advanced models. All models had good accuracy score but their f1 score was limited due to class imbalance between labels. We observed best scores at transformer models as expected and best results was 0.61 macro f1 score from Llama 8B Instruct. Also due to low inter annotator agreement between annotators [1], we can understand these relatively low score results.

VII. INDIVIDUAL CONTRIBUTIONS

A. Aksel Dindisyan

Aksel worked on exploratory data analysis, feature extraction and model development. In Data Analysis part, he generated graphs about features in order to understand data better. In feature extraction part, he developed initial feature extraction that developed by Mustafa by introducing ELFIN and spacy neg for the hedge and negation part respectively. He also speed up the preprocessing and feature extraction by utilising nlp pipelines and parallelization. At model development, he fine tuned a decoder model (LLama 8B Instruct) by using Tinker and used it in classification task.

B. Doruk Yeşil

Doruk focused on transformer-based model development and fine-tuning. He implemented and optimized a RoBERTa-based classifier using paired question&answer inputs, and built the training and evaluation pipeline including dataset splitting for validation, and macro-averaged performance assessment. He conducted hyperparameter optimization to tune learning rate, batch size, number of epochs, and regularization parameters. In addition, he assisted with experiments involving grounding-based and semantic similarity features.

C. İlke Demirkır

Ilke worked on feature extraction and model development. At feature extraction part he improved question polarity, combativeness, and grounding based features by removing basic implementations. At model development, he enhanced encoder models by using grounding documents.

D. Mustafa Selim Türkmen

Mustafa worked on exploratory data analysis, preprocessing, feature extraction and model development. He helped Aksel at data analysis by adding more complex graphs about the database features. Then, he implemented preprocessing using nltk and spacy. After that, he laid the foundation of the feature extraction part by implementing manual methods relied on predefined lists for extracting certain features. At model development part, he worked on neural network training.

E. Sinan Nalbur

Sinan worked on model development part. He trained all the models at the classical baseline such as decision tree, naive bayes, SVM, logistic regression. He also worked on models from advanced machine learning algorithms such as randomforest and xgboost.

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